RAJALAKSHMI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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SEMESTER IV

AL23421 - MACHINE LEARNING LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	Smart Resume Analyzer
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INTRODUCTION

In today's competitive job market, recruiters receive a large number of resumes for each job posting, making manual screening time-consuming and inefficient. Machine Learning and Natural Language Processing (NLP) provide automated ways to analyze textual data and extract meaningful insights.

This project focuses on developing a **Smart Resume Skill Analyzer** that automatically extracts skills from resumes and compares them with job requirements. The system uses NLP techniques to process resume text and applies a rule-based and similarity-based approach to identify relevant skills.

The primary aim of this project is to simplify the recruitment process by providing a **skill match score**, identifying **missing skills**, and helping candidates improve their profiles.

PROBLEM STATEMENT

Manual resume screening is inefficient, prone to bias, and difficult to scale. Recruiters often struggle to quickly identify whether a candidate possesses the required skills.

The problem is to design a system that:

- Automatically extracts skills from resumes
- Compares them with job role requirements
- Calculates a skill match score
- Suggests missing skills

GOAL

The main goals of this project are:

- To build a system that extracts skills from resumes automatically
- To compare candidate skills with predefined job roles
- To calculate a match percentage
- To recommend missing skills to users

Expected outcomes:

- Faster resume screening
- Improved accuracy in skill matching
- Helpful feedback for candidates

THEORETICAL BACKGROUND

1. Natural Language Processing (NLP)

NLP is a field of Artificial Intelligence that enables machines to understand and process human language. In this project, NLP is used to extract meaningful words (skills) from resume text.

2. Text Preprocessing

The resume text undergoes preprocessing steps such as:

- Lowercasing
- Removing punctuation
- Tokenization

3. Skill Extraction

Skills are extracted by matching words from the resume with a predefined skill dataset.

Let:

- $R = \{r_1, r_2, \ldots, r_n\}$be resume words
- $S = \{s_1, s_2, \ldots, s_m\}$be skill dataset

Extracted skills:

Extracted Skills=R∩S

4. Skill Matching Algorithm

Let:

- A= Resume skills
- B= Job required skills

Match Score:

Match Score=/B//A∩B/×100

Literature Survey

- Resume parsing using NLP has been widely used in recruitment systems
- TF-IDF and cosine similarity are commonly used for text comparison
- Rule-based systems are simple and effective for skill extraction

Justification for Choosing Algorithm

- Rule-based skill matching is simple and efficient
- Works well for small datasets
- Easy to implement and understand for a mini project

ALGORITHM EXPLANATION WITH EXAMPLE

Algorithm Used: Skill Extraction and Matching Algorithm

The Smart Resume Skill Analyzer follows a rule-based and comparison-driven approach to extract and evaluate skills from a resume.

Step-by-Step Algorithm

Input:

- Resume file (PDF/DOCX)
- Predefined skill dataset
- Selected job role

Output:

- Extracted skills
- Matched skills
- Missing skills
- Match score (%)

Example

Input Resume Content

Experienced in Python, SQL, and Machine Learning. Worked on data analysis projects.

Skill Dataset

Python, SQL, Machine Learning, Deep Learning, TensorFlow

Extracted Skills

Python, SQL, Machine Learning

Job Role: Data Scientist

Required Skills:

Python, SQL, Machine Learning, Deep Learning

Matched Skills

Python, SQL, Machine Learning

Missing Skills

Deep Learning

Match Score Calculation

Match Score=43×100=75%

Final Output

- Extracted Skills: Python, SQL, Machine Learning
- Matched Skills: Python, SQL, Machine Learning
- Missing Skills: Deep Learning
- Match Score: **75%**

Conclusion of Algorithm

This algorithm efficiently identifies relevant skills from resumes and compares them with job requirements. It provides a simple yet effective way to evaluate candidate suitability and highlight areas for improvement.

IMPLEMENTATION CODE AND DATASET USED

- Dataset Used – https://github.com/JaniseDeepthi/MachineLearning-Miniproject/tree/main/resume-analyzer/dataset

Category	Description / Example
Dataset Name	Skill Dataset
Domain	NLP
Objective	Classification
Data Source	Public (Custom Created)
Number of Samples	~25 skills
Data Type	Text
Instance Description	Each entry is a skill
Features (Inputs)	Skill names
Feature Types	Text
Target Variable	Not Applicable
Number of Classes	Not Applicable
License	Free
Version	1.0

Python code

App.py:

```
from flask import Flask, render_template, request

import os

from resume_parser import extract_resume_text

from skills import extract_skills

from model import match_skills

app = Flask(__name__)

UPLOAD_FOLDER = "uploads"

app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER
```

```
job_roles = {
"Data Scientist": ["python","machine learning","pandas","numpy","sql"],
"Web Developer": ["html","css","javascript","react","nodejs"],
"Data Analyst": ["python","sql","power bi","tableau","data analysis"]}
@app.route("/")
def index():
    return render_template("index.html", roles=job_roles.keys())
@app.route("/analyze", methods=["POST"])
def analyze():
    file = request.files["resume"]
    role = request.form["role"]
    filepath = os.path.join(app.config['UPLOAD_FOLDER'], file.filename)
    file.save(filepath)
    text = extract_resume_text(filepath)
    resume_skills = extract_skills(text)
    job_skills = job_roles[role]
    matched, missing, score = match_skills(resume_skills, job_skills)
    return render_template("result.html",resume_skills=resume_skills, matched=matched,
missing=missing, score=score)
if __name__ == "__main__":
    app.run(debug=True)
```

Resume_Parser.py:

```
import docx2txt
from pdfminer.high_level import extract_text
```

```
def extract_resume_text(file_path):
    if file_path.endswith(".pdf"):
        text = extract_text(file_path)
    elif file_path.endswith(".docx"):
        text = docx2txt.process(file_path)
    else:
        text = ""
    return text.lower()
```

Skill_Extractor.py:

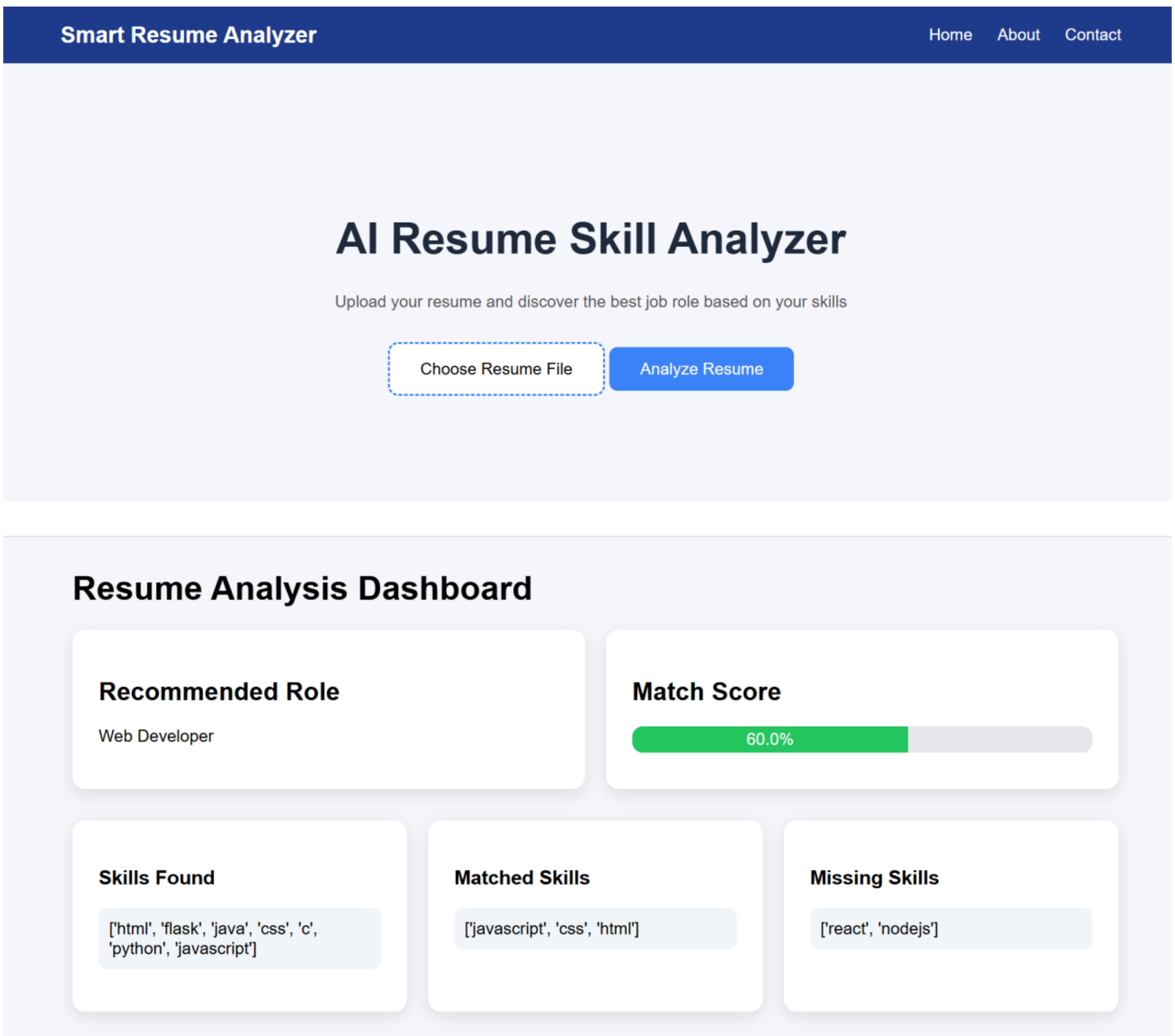
```
import pandas as pd
skills_db = pd.read_csv("dataset/skills.csv")
skills_list = skills_db['skill'].tolist()
def extract_skills(resume_text):
    found_skills = []
    for skill in skills_list:
        if skill in resume_text:
            found_skills.append(skill)
    return found_skills
```

Job_Matcher.py:

```
def match_skills(resume_skills, job_skills):
    matched = []
    missing = []
    for skill in job_skills:
```

```
        if skill in resume_skills:
            matched.append(skill)
        else:
            missing.append(skill)
    score = len(matched) / len(job_skills) * 100
    return matched, missing, score
```

OUTPUT



RESULTS AND FUTURE ENHANCEMENT

Results

- Successfully extracts skills from resumes
- Provides match percentage
- Identifies missing skills

Future Enhancements

- Use advanced NLP models like TF-IDF or BERT
- Add resume ranking system
- Integrate job recommendation system
- Build a React frontend
- Add real-time recruiter dashboard

Git Hub Link of the project along with the report	**https://github.com/JaniseDeepthi/MachineLearning-Miniproject**

REFERENCES

1. Jurafsky & Martin – *Speech and Language Processing*
2. Scikit-learn Documentation
3. NLTK Official Documentation
4. Kaggle datasets (resume/skills datasets)
5. Research papers on resume parsing